

Aurora X1 Field Drone Technical Specification

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1 Overview

The Aurora X1 is a rugged quadcopter designed for field inspection of power lines, pipelines and wind turbines. It is built for crews that need to launch within two minutes of arriving on site and keep flying in light rain and gusty wind. The X1 replaces the Aurora X0, which was discontinued in 2024.

Target customers are utility companies and infrastructure inspection contractors. The X1 is sold as a kit that includes the aircraft, two batteries, the Skylark GC-3 ground controller and a hard-shell transport case.

2 Hardware

2.1 Airframe

The airframe is a carbon-fibre monocoque with folding arms. Unfolded, the diagonal wheelbase is 780 mm; folded, the aircraft measures 410 x 290 x 180 mm. Take-off weight is 4.2 kg with one battery and no payload, and the maximum take-off weight is 6.4 kg.

2.2 Propulsion

Four brushless outrunner motors drive 18-inch folding propellers. Each motor has its own electronic speed controller with thermal protection. If a single motor fails above 15 metres, the X1 can still complete a controlled landing on the remaining three motors in a degraded yaw mode.

2.3 Battery System

The X1 uses hot-swappable smart batteries. Swapping a battery takes under 30 seconds and does not require powering down the flight computer.

Property	Value
Chemistry	Lithium-ion (21700 cells)
Capacity	12,000 mAh
Nominal voltage	44.4 V (12S)
Energy	533 Wh
Charge time (0-100%)	75 minutes with the CH-400 charger
Rated cycles	400 cycles to 80% capacity
Battery weight	1.9 kg

Batteries discharge themselves to 60% after 10 days without use to extend their service life. For storage longer than three months, keep batteries between 15 and 25 degrees Celsius.

3 Performance

Metric	Value
Maximum flight time (no payload)	46 minutes
Flight time with 2 kg payload	38 minutes
Maximum range (line of sight)	12 km
Maximum horizontal speed	72 km/h
Maximum wind resistance	12 m/s
Service ceiling	5,000 m above sea level
Operating temperature	-20 to 50 degrees Celsius
Ingress protection	IP55

Flight times are measured at sea level and 25 degrees Celsius in still air, landing with 10% battery remaining. Expect flights to be roughly 15% shorter below 0 degrees Celsius.

4 Sensors and Payload

The standard payload is the Skylark VZ-20 gimbal camera with a 20-megapixel 1-inch sensor and 12x hybrid zoom. The optional TH-640 thermal module adds a 640 x 512 radiometric thermal sensor for detecting hotspots on electrical equipment. The payload mount uses the Skylark QuickLock interface and supports payloads of up to 2.2 kg.

Obstacle sensing uses six stereo vision pairs plus an upward-facing time-of-flight sensor, giving omnidirectional detection from 0.5 m to 40 m. RTK positioning is available with the optional RTK-2 module, which provides 1 cm + 1 ppm horizontal accuracy.

5 Safety Features

- Automatic return-to-home when the battery reaches the calculated reserve, or when the control link is lost for more than 3 seconds.
- Geofencing with a configurable altitude ceiling (default 120 m).
- ADS-B receiver that warns the pilot about crewed aircraft within 10 km.
- Emergency propeller stop, triggered by holding both control sticks inward for 2 seconds.
- Parachute port compatible with the SkySafe P2 parachute system.

6 Maintenance

Following the maintenance schedule is a condition of the warranty.

Interval	Task
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Before every flight	Inspect propellers for cracks; check battery latches; verify compass calibration
Every 50 flight hours	Replace propellers; clean motor vents; update firmware
Every 200 flight hours	Replace motor bearings; inspect arm hinges; factory IMU calibration
Every 12 months	Full service at an authorised Skylark service centre

7 Warranty

The Aurora X1 carries a 12-month limited warranty covering manufacturing defects. Batteries are covered for 6 months or 150 cycles, whichever comes first. Crash damage is not covered, but customers can buy Skylark Care Plus, which covers up to two accidental-damage replacements per year for an annual fee of 890 USD.